

Retrieval-Augmented Generation (RAG) - Study Notes

What is RAG?

Retrieval-Augmented Generation (RAG) is a technique that gives a large language model relevant information from external documents before it answers. Instead of relying only on what it learned during training, the model reads retrieved text and uses it as context. RAG was introduced by Patrick Lewis and colleagues at Facebook AI Research in a 2020 paper.

The three stages of RAG

1. Indexing: documents are split into chunks, each chunk is converted into an embedding vector, and the vectors are stored in a vector index such as FAISS.
2. Retrieval: the user question is embedded with the same model and the most similar chunks are found using similarity search.
3. Generation: the retrieved chunks are added to the prompt and the LLM writes an answer grounded in that context.

Why use RAG?

RAG lets a model answer questions about private or recent documents without retraining, reduces hallucinations, and makes it possible to show the sources that an answer was based on.

What is FAISS?

FAISS (Facebook AI Similarity Search) is an open-source library developed by Meta for efficient similarity search over dense vectors. An IndexFlatIP index compares a query vector with every stored vector using the inner product.